Department of Computer Science and Engineering

Course Code: CSE 370	Credits: 3.0
Course Name: Database Systems	Semester: Fall 2025

Lab 02: SQL Subqueries & Aggregate Functions

Activity List

- **All commands are shown in the red boxes.**
- **In the green box, write the appropriate query/answer.**
- **All new queries should be typed in the command window after mysql>**
- **Start by connecting to the server using: mysql -u root -p [password: <just press enter>]**
- **For more MySQL queries, go to www.w3schools.com/sql or google it!**

Initial Table: It's a bit different than Lab 01!

std_id	name	major	section	days_present	project_marks	cgpa	submission_date
s001	Abir	CS	1	10	18.5	3.91	2018-09-15
s002	Nafis	CSE	1	12	20	3.86	2018-08-15
s003	Tasneem	CS	1	8	18	3.57	2018-09-18
s004	Nahid	ECE	2	7	16.5	3.25	2018-08-20
s005	Arafat	CS	2	11	20	4.0	2018-09-13
s006	Tasneem	CSE	1	12	17.5	3.7	2018-08-15
s007	Muhtadi	ECE	1	10	19	3.67	2018-09-16

Link for Table Data: https://docs.google.com/document/d/1j6zmKf3cUr7zQKNxByfRPh0_AaTUn3hTyW6_AKuCFQA/

The purpose of the SELECT statement is to retrieve and display data from one or more database tables. It is an extremely powerful command. SELECT is the most frequently used SQL command and has the following general form:

SELECT [DISTINCT | ALL] {* | [columnExpression [AS newName]] [, . . .]}

FROM TableName [alias] [, . . .]

[WHERE condition]

[GROUP BY columnList] [HAVING condition]

[ORDER BY columnList]

columnExpression represents a column name or an expression, TableName is the name of an existing database table or view that you have access to, and alias is an optional abbreviation for TableName.

The sequence of processing in a SELECT statement is:

↓	**FROM** specifies the table or tables to be used
	WHERE filters the rows subject to some condition
	GROUP BY forms groups of rows with the same column value
	HAVING filters the groups subject to some condition
	SELECT specifies which columns are to appear in the output
	ORDER BY specifies the order of the output

The order of the clauses in the SELECT statement cannot be changed. The only two mandatory clauses are the first two: SELECT and FROM; the remainder are optional. The SELECT operation is closed: the result of a query on a table is another table.

<u>Task 1: Aggregate Functions, Group By and Having:</u>

Retrieve the minimum CGPA/Project_marks from the table	SELECT MIN(***cgpa***) FROM ***Lab_Grades***;
Retrieve the total number of students and the average projects marks	SELECT COUNT(*) as ***total_students***, AVG(***project_marks***) as ***average_project_marks*** FROM ***Lab_Grades;***
Find the sum of the number of days present.	SELECT SUM(***days_present***) FROM ***Lab_Grades;***

- How will you retrieve the last submission date?

Find Minimum and Maximum CGPA of each major	SELECT ***major, MIN***(***CGPA***) as ***minCGPA,*** MAX(***CGPA)*** as ***maxCGPA*** *FROM* ***Lab_Grades*** GROUP BY ***major;***
Retrieve total number of students for each major	SELECT ***major,*** COUNT***(*)*** FROM ***Lab_Grades*** GROUP BY ***major;***

- What is the purpose of the group by keyword? In the above command, if we group by sub_date, instead of major, what will be the output?

For each major find the minimum and maximum CGPA/Project_marks, but only if there were at least 2 students in the major

```
SELECT major, MIN(cgpa) as minCGPA, MAX(cgpa) as maxCGPA
FROM Lab_Grades GROUP BY major HAVING COUNT(*)>=2;
```

For each major find the minimum and maximum CGPA/Project_marks, but consider only students who submitted before or on 15th sep

```
SELECT major, MIN(cgpa) as minCGPA, max(cgpa) as maxCGPA FROM Lab_Grades
WHERE submission_date<='2018-09-15' GROUP BY major;
```

- The having and where clauses are both used to specify a condition when selecting rows. What is the difference between them?

Task 2: Sub Queries/Nested Queries, Any and All:

- Think about how you can retrieve the names of students who got the highest project marks. Try out your query. Did you get the "correct" response according to the table?

Now, try the nested/sub query on the right

```
SELECT name FROM Lab_Grades
WHERE project_marks=(SELECT MAX(project_marks)
                     FROM Lab_Grades);
```

Retrieve the CSE students whose CGPA/Project_marks is higher than at least 1 CS students

```
SELECT * from Lab_Grades WHERE major = 'CSE' and cgpa>ANY
(SELECT cgpa FROM Lab_Grades WHERE major = 'CS');
```

Retrieve the CSE students whose CGPA/Project_marks is higher than all CS students

```
SELECT * FROM Lab_Grades WHERE major = 'CSE' and cgpa>ALL
(SELECT cgpa FROM Lab_Grades WHERE major = 'CS');
```

- Did you understand the role of "any" and "all" in the above queries? Explain below.

- Retrieve the names of the students who have received marks greater than at least 1 student doing the same major as them.[Hint: see next command]

Task 3: Correlated Subqueries and Exists:

Select those majors for which at least 1 student has CGPA lower than 3.7

SELECT DISTINCT ***major*** FROM ***Lab_Grades L1*** WHERE EXISTS (SELECT * FROM ***Lab_Grades L2*** WHERE ***L2.major=L1.major*** and ***L2.cgpa<3.7***);

- L1 and L2 are temporary aliases and create two separate instances for Lab_Grades; why are they required?

Retrieve the name of student who has obtained maximum marks in project using exists

SELECT **name** FROM **Lab_Grades L1** WHERE NOT EXISTS (SELECT * FROM **Lab_Grades L2** WHERE **L2.std_id!=L1.std_id** AND **L2.project_marks>L1.project_marks**);

Retrieve the name of student who has obtained maximum marks in project and who is unique using exists

SELECT **name** FROM **Lab_Grades L1** WHERE NOT EXISTS (SELECT * FROM **Lab_Grades L2** where **L2.std_id!=L1.std_id** AND **L2.project_marks>=L1.project_marks**);

- Please identify the difference between the above two queries. [Hint: 1 asks for unique-only 1 student got the highest and the other didn't]

Retrieve the total number of students who obtained the maximum marks. There are many ways of solving one task; a few ways for this one are shown below.

SELECT COUNT(*) FROM **Lab_Grades L1** WHERE NOT EXISTS (SELECT * FROM **Lab_Grades L2** WHERE **L2.std_id!=L1.std_id** and **L2.project_marks>L1.project_marks**);

SELECT COUNT(*) FROM ***Lab_Grades*** WHERE ***project_marks*** = (SELECT MAX(***project_marks)*** FROM ***Lab_Grades***);

SELECT COUNT(*) FROM **Lab_Grades** WHERE **project_marks** >=ALL (SELECT project_marks FROM Lab_Grades);

Retrieve the major which has the highest number of students enrolled.

```
SELECT major FROM Lab_Grades GROUP BY major HAVING count(*)
>= ALL (SELECT count(*) FROM Lab_Grades GROUP BY major);
```

Task 4: Take a Quiz

Go to https://sqlzoo.net/wiki/Nested_SELECT_Quiz to test your understanding of the queries taught in class.